Deep Learning for Extreme Weather Forecasting

Abstract

The prediction of extreme weather events—encompassing tropical cyclones, severe convective storms, prolonged droughts, and anomalous heatwaves—remains one of the most critical and computationally demanding challenges in atmospheric science. Historically, operational meteorological forecasting has depended on Numerical Weather Prediction (NWP) models, which utilize computationally intensive numerical simulations of fluid dynamics and thermodynamics. Recently, data-driven Machine Learning Weather Prediction (MLWP) systems have emerged as a transformative alternative, demonstrating the capability to match or exceed the accuracy of traditional NWP models at a fraction of the computational cost. However, because foundational deep learning models are typically optimized for average performance metrics, they frequently suffer from a "regression to the mean," smoothing out fine-grained spatio-temporal details and systematically underestimating heavy-tailed extreme events. This comprehensive survey provides an exhaustive review of the state-of-the-art in deep learning architectures tailored for extreme weather forecasting. The analysis categorizes the methodological landscape into deterministic global foundation models, probabilistic and generative frameworks, physics-AI hybrid systems, and high-resolution downscaling techniques. Furthermore, it examines the application of these architectures to specific high-impact meteorological phenomena. Finally, the survey delineates critical ongoing challenges—namely, the extrapolation deficit for record-breaking events, the necessity for extreme-value loss functions, the evolution of operational benchmarking from reanalysis to in-situ observation, and vulnerabilities related to adversarial attacks—and proposes pathways for future research in trustworthy and resilient meteorological artificial intelligence.

Introduction

The increasing frequency and intensity of extreme weather events, driven by anthropogenic climate change, pose severe threats to global ecosystems, critical infrastructure, and human life [1]. Accurate and timely weather forecasting is essential for disaster preparedness, agricultural planning, aviation safety, and energy management [2, 3]. Traditionally, Numerical Weather Prediction (NWP) models, such as the European Centre for Medium-Range Weather Forecasts (ECMWF) Integrated Forecasting System (IFS) and the National Oceanic and Atmospheric Administration (NOAA) Global Forecast System (GFS), have served as the cornerstone of operational meteorology [4]. NWP systems simulate the Earth's atmosphere by solving complex primitive equations of fluid dynamics and thermodynamics on a three-dimensional grid. Despite their historical success, NWP models exhibit notable limitations: they are computationally exorbitant, require significant integration time before issuing forecasts, and rely heavily on parameterized approximations for sub-grid scale processes—such as cloud microphysics, convection, and radiation [5, 6]. These parameterizations introduce persistent systematic biases, particularly in predicting extreme precipitation and localized severe weather phenomena [7, 8].

In recent years, the paradigm has shifted significantly toward Machine Learning Weather Prediction (MLWP) [9]. Deep learning architectures—specifically Convolutional Neural Networks (CNNs), Graph Neural Networks (GNNs), and Vision Transformers (ViTs)—have demonstrated unprecedented capabilities in modeling the non-linear dynamics of the Earth system directly from historical reanalysis data [10, 11]. Data-driven global models can now generate highly accurate medium-range forecasts in a matter of seconds on a single GPU or TPU, representing a computational speedup of up to four orders of magnitude compared to traditional supercomputer-bound NWP runs [12, 13, 14]. This "quiet revolution" in meteorology has led major forecasting centers to begin integrating AI models alongside traditional physics-based operational suites [15].

Despite these remarkable advances, the application of deep learning to extreme weather forecasting introduces distinct theoretical and practical hurdles. Deep learning models, by their statistical nature, excel at interpolation within dense data distributions but frequently struggle with extrapolation—specifically, predicting rare, out-of-distribution, or record-breaking tail events [16, 17]. Models optimized via standard objective functions like Mean Squared Error (MSE) or Mean Absolute Error (MAE) tend to generate physically smoothed fields at longer lead times. This inherent suppression of variance eliminates the sharp atmospheric gradients that characterize extreme hazards like rapidly intensifying hurricanes or flash floods [18, 19, 20]. Consequently, while global AI models achieve superior scores on standard metrics, they often fail to capture the absolute intensity of catastrophic weather events.

This survey systematically reviews the intersection of deep learning and extreme weather forecasting. It structures the rapidly evolving field into distinct architectural methodologies, evaluates their efficacy across discrete extreme weather phenomena, and critically analyzes the persistent challenges hindering the full operational deployment of AI-based early warning systems. By synthesizing recent breakthroughs in generative AI, hybrid modeling, and specialized loss functions, this manuscript provides a roadmap for the next generation of resilient meteorological AI.

Methods

The landscape of deep learning for weather forecasting can be broadly categorized into four primary paradigms: deterministic global foundation models, probabilistic and generative ensembles, physics-AI hybrid frameworks, and high-resolution downscaling/nowcasting architectures.

Deterministic Global Foundation Models

The first wave of highly successful MLWP models focused on deterministic, autoregressive global forecasting. These models are typically trained on decades of the ECMWF ERA5 reanalysis dataset to predict the atmospheric state at time $t + \Delta t$ given the state at time t, iterating this process to achieve medium-range (1-15 day) forecasts [10, 12, 21].

Pangu-Weather: Developed by Huawei, Pangu-Weather utilizes a 3D Earth-Specific

Transformer (3DEST) architecture designed to process complex, non-uniform 3D meteorological data spanning multiple pressure levels [12]. To mitigate the accumulation of iteration errors common in autoregressive models, Pangu-Weather employs a hierarchical temporal aggregation strategy with multiple sub-models trained for specific temporal steps (e.g., 1-hour, 3-hour, 6-hour, and 24-hour intervals) [12, 22]. Pangu-Weather demonstrated higher precision than traditional numerical methods for global deterministic metrics and successfully tracked the trajectory of complex systems like Typhoon Mawar [12]. However, independent analyses indicate that while Pangu-Weather excels at geopotential and temperature tracking, it struggles with the absolute intensity of extreme events, such as peak cyclonic wind speeds, due to the spatial smoothing inherent in deterministic MSE training [23, 24].

GraphCast: Google DeepMind's GraphCast approaches the atmosphere as a complex network, employing Graph Neural Networks (GNNs) mapped to an icosahedral multi-mesh representation of the Earth [13, 25]. Operating at a 0.25° resolution, GraphCast predicts the next 6-hour state of the atmosphere. By minimizing localized message-passing bottlenecks and circumventing the pole-singularity issues of standard latitude-longitude grids, GraphCast has consistently outperformed the operational IFS-HRES on a vast majority of verification targets. It has shown superior capability in tracking severe synoptic-scale events such as atmospheric rivers and extreme temperature anomalies, though it shares the deterministic limitation of underpredicting extreme precipitation intensities [13, 25].

FengWu and FengWu-GHR: Developed by the Shanghai AI Laboratory, the FengWu family integrates multi-modal and multi-task deep learning [26]. Its advanced iteration, FengWu-GHR, achieved a significant milestone by operating at a globally unprecedented 9 km resolution (2001×4000 grid), aligning with the highest resolution of operational NWPs [27]. FengWu utilizes a novel decompositional and combinational transfer learning approach, coupled with low-rank adaptation, to extrapolate from low-resolution pretraining to high-resolution deployment. This significantly enhances the granularity of extreme heatwave and winter storm predictions [26, 27]. Furthermore, FengWu has been integrated with 4D-Variational (4DVar) data assimilation (FengWu-Adas) to create a purely AI-driven cyclic forecasting system that directly ingests sparse observational data without relying on NWP analysis fields for initialization [28].

Aurora: Microsoft Research introduced Aurora, a 1.3 billion parameter 3D Swin Transformer foundation model. Aurora distinguishes itself through its pretraining regimen: rather than relying solely on ERA5, it is pretrained on a diverse multimodal corpus of over one million hours of data, including climate simulations, GFS analysis, and IFS high-resolution forecasts [29, 30]. Through a Modality-Guided Multi-head Self-Attention mechanism, Aurora adapts to heterogeneous variables and extreme events, demonstrating state-of-the-art short-range skill (1-7 days) for tropical cyclones and atmospheric rivers [31]. However, empirical evaluations indicate that its predictive amplitude for extreme surface impacts regresses toward climatology beyond 10 days, conforming to theoretical Lorenz predictability limits for deterministic systems [29, 32].

AIFS: Marking the formal adoption of MLWP by traditional meteorological institutions, the

ECMWF developed the Artificial Intelligence Forecasting System (AIFS). AIFS utilizes an encoder-processor-decoder architecture where the encoder and decoder are attention-based GNNs, and the processor is a sliding-window transformer [15, 33]. Running operationally alongside the physics-based IFS, AIFS has demonstrated parity or superiority in metrics such as the Anomaly Correlation Coefficient (ACC) and Root Mean Square Error (RMSE), producing forecasts ten times faster and consuming orders of magnitude less energy than its numerical counterpart [34, 35].

Model	**Core Architecture**	**Organization**	**Key Architectural Feature**
Pangu-Weather	3D Earth-Specific Transformer	Huawei	Multi-timescale hierarchical sub-models
GraphCast	Graph Neural Network	Google DeepMind	Icosahedral multi-mesh message passing
FengWu-GHR	Multi-task Deep Learning	Shanghai AI Lab	9 km ultra-high resolution transfer learning
Aurora	3D Swin Transformer	Microsoft Research	Multi-dataset multimodal pretraining
AIFS	GNN + Sliding Window Transformer	ECMWF	Operational integration alongside NWP

Probabilistic and Generative Models

While deterministic models effectively minimize average error, they inherently produce blurred forecasts when atmospheric uncertainty is high, effectively suppressing the prediction of extreme, low-probability events [18, 36]. To address this, the field has rapidly moved toward probabilistic and generative AI, which models the full predictive distribution to quantify uncertainty and generate highly detailed, physically realistic ensemble members [37, 38].
Diffusion Models (GenCast and FGN): Google DeepMind developed GenCast, a conditional diffusion model capable of generating a 15-day global forecast ensemble in approximately 8 minutes [39]. Rather than predicting a single deterministic state, GenCast models the

conditional joint probability distribution of the atmosphere. By denoising random latent inputs conditioned on previous atmospheric states via a probability flow Ordinary Differential Equation (ODE), GenCast produces 50 distinct ensemble members [39, 40]. Crucially, GenCast optimizes for the Continuous Ranked Probability Score (CRPS), which evaluates both calibration and accuracy. Consequently, it maintains sharp atmospheric gradients and predicts extreme weather, tropical cyclones, and wind power generation better than the ECMWF ENS system [39, 41]. DeepMind also proposed FGN, which models both epistemic and aleatoric uncertainty to generate ensembles with learned model-perturbations [41].

Evidential Deep Learning (EDL): Because generating massive ensembles via diffusion or Monte Carlo dropout still carries computational overhead, Evidential Deep Learning (EDL) has been proposed as a highly efficient alternative. EDL treats learning as an evidence acquisition process, estimating predictive uncertainty directly during the training phase without the need for repetitive sampling [2]. This allows a single deterministic forward pass to output both the prediction and a calibrated confidence interval, drastically reducing the computational footprint required for real-time climate risk assessments [2, 42].

Physics-AI Hybrid Frameworks

Purely data-driven models face skepticism from the meteorological community regarding their adherence to fundamental conservation laws (mass, momentum, energy) and their ability to extrapolate under shifting climate baselines [16, 43]. Physics-AI hybrid frameworks seek to combine the reliability and out-of-distribution robustness of physical laws with the representational power and computational efficiency of deep learning.

NeuralGCM: Google Research developed NeuralGCM, an open-source hybrid atmospheric model. NeuralGCM retains a traditional, differentiable fluid dynamics solver (a dynamical core) for large-scale, resolved dynamical processes, ensuring the conservation of mass and momentum [5, 44]. However, it replaces traditional sub-grid parameterizations—heuristic formulas used for clouds, radiation, and precipitation—with neural networks [5, 45]. Crucially, the machine-learning components of NeuralGCM are trained directly on high-fidelity observational data, such as NASA's Global Precipitation Measurement (GPM) satellite observations, rather than relying solely on reanalysis data [5, 46]. NeuralGCM demonstrates extreme stability for long-term climate simulations in warmer climates and achieves substantial error reductions for the most intense 0.1% of rainfall events, rectifying the tendency of pure physical models to overpredict light rain and underpredict heavy downpours [5, 47, 48].

High-Resolution Downscaling and Nowcasting

Nowcasting (0–6 hour forecasting) and downscaling are critical for predicting severe, highly localized events like tornadoes, hail, and flash floods. These phenomena occur at spatial and temporal scales too fine for global medium-range models to resolve [49, 50].

NowcastNet and DGMR: Deep Generative Models of Radar (DGMR) and NowcastNet treat precipitation nowcasting as an advanced video generation task [51, 52]. NowcastNet is a non-linear model unifying physical-evolution schemes with conditional Generative Adversarial Networks (GANs). It utilizes a deterministic evolution network to model advective conservation and a stochastic generative network to predict short-lived convective details from latent

Gaussian vectors. This allows NowcastNet to maintain sharp multi-scale patterns at a 1-2 km resolution for up to 3 hours, avoiding the unphysical blurring typically seen in standard U-Net architectures optimized via MSE [51, 53].
Global MetNet: Traditional nowcasting relies heavily on ground-based Doppler radar, which is dense in the Global North but virtually nonexistent in the Global South and over oceans [54]. Global MetNet bridges this disparity by utilizing a deep neural network that ingests geostationary satellite data, GPM CORRA datasets, and global NWP inputs to produce 15-minute, 5 km resolution precipitation forecasts up to 12 hours ahead, entirely bypassing the need for ground radar [54].
CorrDiff and StormCast: Downscaling coarse climate projections to high-resolution local impacts is traditionally achieved via dynamic NWP nesting, which is computationally prohibitive [55]. NVIDIA's CorrDiff utilizes a two-step generative approach: a U-Net first predicts a deterministic mean, followed by a diffusion model that learns to add high-frequency stochastic corrections (akin to Reynolds decomposition in fluid dynamics) [56, 57]. CorrDiff successfully downscales 25 km global data to 2 km regional data, capturing extreme wind and rainfall bands near typhoon eyewalls at 1,300 times the energy efficiency of a comparable WRF numerical simulation [56, 58, 59]. Similarly, StormCast utilizes autoregressive diffusion to emulate 3 km convection-allowing models (CAMs), predicting severe thunderstorms and mesoscale convective systems [60].

Applications to Specific Extreme Weather Events

Tropical Cyclones and Rapid Intensification

Tropical cyclones (TCs) are among the most destructive natural hazards, and their accurate forecasting depends on predicting two distinct variables: track (spatial trajectory) and intensity (maximum sustained wind speed and minimum sea-level pressure) [61].
Track and Intensity Forecasting: Deterministic global models often struggle with TC intensity because extrema are inherently smoothed out at the standard 0.25° grid resolution [62]. To counteract this, models like BaguanCyclone introduce probabilistic center refinement modules. Instead of snapping TC centers to a discrete grid, BaguanCyclone maps the cyclone to a continuous probability density field via truncated Gaussian kernels. This enables sub-grid track precision and region-aware intensity extraction, significantly reducing discretization errors [63]. Other approaches employ Transformer variants (e.g., iTransformer, TVFormer), which are adept at capturing the intrinsic non-linear temporal dependencies of historical TC tracks, drastically reducing longitude and latitude track errors compared to recurrent neural networks (RNNs) [64, 65]. Furthermore, hierarchical post-processing networks applied to Neural Weather Models (NeWMs) have shown that while global AI models underestimate raw hurricane intensity, their output contains the necessary environmental context to accurately diagnose TC intensity when fed into a secondary deep learning post-processor up to five days ahead [62, 66].
Rapid Intensification (RI): Rapid Intensification—defined as an increase in maximum sustained winds of at least 30 knots within a 24-hour period—is a complex thermodynamic phenomenon notoriously difficult for traditional statistical-dynamical models like SHIPS (Statistical Hurricane

Intensification Predictive Scheme) to predict [67, 68]. AI models utilizing deep Convolutional LSTMs (ConvLSTM) have proven superior by simultaneously encoding human-defined environmental predictors and automatically extracting asymmetric convective features (e.g., eye formation, central dense overcast patterns) directly from multi-channel satellite imagery [69, 70]. Models evaluating the Net Energy Gain Rate (NGR) index through Multi-Layer Perceptrons (MLPs) have yielded higher Probability of Detection (POD) and lower False Alarm Ratios (FAR) for RI events compared to operational baselines, demonstrating the capacity of deep learning to capture the rapid energy exchanges between the ocean and the atmosphere [71, 72].

Extreme Precipitation and Convective Storms

Severe convective weather (SCW)—including thunderstorms, hail, and extreme precipitation—develops rapidly and exhibits highly localized non-linear dynamics [73]. Conventional optical flow methods for radar extrapolation struggle with the non-linear growth and decay of convective cells, leading to rapid skill degradation [52].

To address this, advanced architectures like the Residual Attention U-Net (RA-UNet) and Pixel-CRN integrate spatial convolution subnets into recurrent units. These models fuse 3D volumetric radar data with NWP reanalysis backgrounds, allowing the network to incorporate broader thermodynamic context (e.g., convective available potential energy, wind shear) rather than relying purely on radar kinematics [52, 74, 75]. In tropical regions like Southeast Asia, where squalls and monsoons dominate, deep learning models utilizing Himawari-8 satellite radiance data and Generative Adversarial Networks (GANs) have been successfully operationalized. These models reconstruct radar-equivalent reflectivity from infrared signatures, providing seamless 8-hour nowcasts of tropical precipitation in data-sparse regions [76, 77, 78]. Additionally, machine learning anomaly detectors, such as the TACLS system, analyze tropospheric water vapor delays in Global Navigation Satellite System (GNSS) signals, successfully predicting imminent flash floods with a 93% accuracy rate [79].

Heatwaves, Droughts, and Subseasonal Anomalies

Prolonged climatic extremes, such as droughts and heatwaves, are governed by complex land-atmosphere feedbacks and large-scale teleconnections (e.g., El Niño-Southern Oscillation (ENSO), Madden-Julian Oscillation (MJO)).

Flash Droughts: Flash droughts emerge rapidly due to acute precipitation deficits paired with anomalously high evapotranspiration. Standard dynamical forecast systems rarely predict flash droughts beyond a 10-day lead time [80]. The Interpretable Transformer for Drought (IT-Drought) overcomes this limitation by forecasting root-zone soil moisture up to 42 days ahead. Utilizing Integrated Gradients (IG) for explainability, IT-Drought quantifies the heterogeneous memory effects of soil moisture and the specific temporal lag impacts of radiation and temperature, providing novel data-driven interpretability for drought onset mechanisms [80, 81]. Other studies utilize XGBoost and LSTMs to forecast the U.S. Drought Monitor (USDM) classifications and directly predict socio-ecological drought impacts (e.g., agricultural losses and fire risks) using indices like the Evaporative Stress Index (ESI) [82, 83, 84].

Subseasonal to Seasonal (S2S) and Monsoons: Forecasting the South Asian and Indian

Summer Monsoons is notoriously challenging, particularly in the 10- to 30-day "predictability gap" [85, 86]. Deep learning has demonstrated advanced capability in predicting the Monsoon Intraseasonal Oscillation (MISO) by projecting dynamical ensemble forecasts onto a lower-dimensional data-driven subspace, drastically improving temporal rainfall correlation over the subcontinent [85]. Similarly, ConvLSTM networks predicting sea level pressure anomalies have proven to be highly effective proxies for forecasting the active-break cycles of the Indian Summer Monsoon at synoptic scales [87]. For large-scale oceanic anomalies, deep learning models utilizing multi-step temporal convolutions with attention mechanisms have successfully extended the prediction lead time of the South Indian Ocean Dipole (SIOD) from 3 months to 7 months, capturing both local wind feedback and remote ENSO teleconnections [88].

Winter Storms and Cold Waves

Winter storms and blizzards cause severe disruptions to infrastructure, yet their mesoscale features (e.g., lake-effect snowbands) possess limited predictability [89]. Researchers have applied LSTMs utilizing bivariate attributes (wind speed and barometric pressure) from polar regions to successfully recognize temporal blizzard patterns [90]. Post-event analysis, critical for disaster recovery and ecological monitoring, has also benefited from AI; deep learning semantic segmentation models like DeepLabv3+ and U-Net have been utilized on multi-spectral satellite imagery to perform rapid, high-accuracy detection of windthrow (uprooted forests) in the immediate aftermath of severe winter storms, vastly outperforming traditional survey methods [91].

Challenges and Prospects

While MLWP has achieved remarkable milestones, predicting the absolute extremes of the Earth system reveals fundamental methodological and operational bottlenecks that require targeted innovation.

The Extrapolation Deficit for Record-Breaking Extremes

A foundational criticism of purely data-driven weather models is their theoretical inability to extrapolate beyond their training domain [16, 92]. Because AI systems learn to map historical precedents, they are inherently biased toward climatological "normality." A comprehensive evaluation of 2020 record-breaking extreme events demonstrated that while models like GraphCast and FuXi outperform the ECMWF HRES on overall spatial metrics, the physics-based HRES consistently outperformed all AI models when predicting unprecedented, record-breaking temperature and wind speed extremes [17, 92, 93]. The AI models systematically underestimated the magnitude of these novel tail events, often smoothing them into less severe categories. Consequently, researchers advocate for the continued parallel use of physics-based models alongside AI, as physical models rely on invariant thermodynamic laws that hold true even when the atmosphere moves into unobserved, climate-shifted states [16, 92].

Loss Function Innovations for Heavy-Tailed Distributions

The standard optimization paradigm for deep learning—minimizing MSE or MAE—is intrinsically detrimental to extreme event forecasting. These loss functions penalize "double errors" heavily (e.g., predicting a storm in slightly the wrong location incurs a penalty for the false alarm and a penalty for the miss), forcing the model to output a blurred, low-intensity consensus of possible states to minimize mathematical risk [94, 95].

To counter class imbalance and heavy-tailed distributions, advanced loss functions are actively being developed. **Extreme Value Loss (EVL)** and tail-weighted loss functions have been introduced to explicitly emphasize rare, high-impact events [94, 96]. For example, in precipitation forecasting, the remainder series containing extreme variations can be fitted with a Generalized Extreme Value (GEV) distribution. LSTMs are then trained to predict the Cumulative Distribution Function (CDF) of this GEV using a tail-weighted loss, successfully transforming extreme prediction into a bounded probabilistic task [96, 97]. Similarly, weighted loss functions targeting the minority class (e.g., strong El Niño/La Niña years) have significantly improved ENSO predictions by artificially reducing the weight of high-frequency normal events during backpropagation [98]. In computer vision tasks for segmenting atmospheric rivers and TCs, optimizing for recall through weighted focal losses prevents the dangerous under-counting of critical extremes [99].

Furthermore, the **Targeted Concept Tuning (TaCT)** framework introduces concept-level plasticity. In standard fine-tuning, forcing an AI model to learn a rare extreme event often overwrites existing representations, leading to "catastrophic forgetting" of general weather patterns [100]. TaCT utilizes continuous counterfactual reasoning to isolate and fine-tune only the specific neurons (concepts) responsible for extreme event failures, allowing surgical correction of typhoon forecasts without degrading the model's baseline performance on standard variables [100].

Next-Generation Benchmarking: From Reanalysis to Reality

The rapid proliferation of MLWP necessitates rigorous, standardized evaluation. **WeatherBench 2** established the current gold standard, providing cloud-optimized ERA5 ground-truth datasets and specific metrics to evaluate global models in a like-for-like fashion [101, 102, 103].

However, WeatherBench 2 relies on ERA5 reanalysis data, which is itself a product of numerical modeling. This means ERA5 carries the inherent biases of the IFS assimilation system and does not perfectly reflect real-time operational constraints or the true state of the atmosphere at ground level [104, 105]. To address this, **RealBench** (and related datasets like **WeatherReal**) evaluates AI models strictly against real-time operational analysis and in-situ data from over 10,000 global weather stations. This eliminates data leakage and exposes significant real-world performance gaps in AI models that otherwise perform flawlessly on ERA5 [104, 105].

Concurrently, **ExtremeWeatherBench (ExEBench)** and **TCBench** have been introduced as targeted frameworks that evaluate Foundation Models exclusively on high-impact events, assessing the zero-shot transferability and physical consistency of AI under heavy-tailed noise [106, 107, 108].

Benchmark Framework	Primary Ground Truth	Key Focus Area	Reference

WeatherBench 2	ERA5 Reanalysis	Global medium-range deterministic & probabilistic forecasting	[101, 103]
RealBench / WeatherReal	In-situ Station Observations	Zero-leakage operational evaluation, true surface conditions	[104, 105]
ExEBench	Multi-source EO & Weather	Generalizability across floods, wildfires, heatwaves	[106, 107]
TCBench	IBTrACS	Tropical Cyclone track and intensity	[108]

Interpretability, Security, and Trustworthy AI

The operational deployment of MLWP requires immense trust from meteorologists, necessitating advancements in eXplainable AI (XAI). While physical models are inherently transparent—errors can be traced back to specific parameterized equations of fluid dynamics—data-driven models act as black boxes [109]. Tools like Layer-wise Relevance Propagation (LRP) and Integrated Gradients (IG) are increasingly being utilized to map model outputs back to physical input features, verifying that the AI is learning geophysically valid relationships (e.g., wind shear driving storm intensity) rather than spurious statistical correlations [110, 111].

Additionally, the reliance on aggregated, multi-source observational data introduces a novel threat vector: adversarial vulnerability. Research has demonstrated that autoregressive diffusion models (like GenCast) can be manipulated via subtle, adversarial perturbations to local sensor data. These malicious inputs can theoretically force the AI to fabricate non-existent extreme events (e.g., phantom hurricanes) or conceal legitimate hazards, highlighting an urgent need for robust adversarial defense mechanisms within operational AI forecasting pipelines [109, 112].

Conclusion

The application of deep learning to extreme weather forecasting has catalyzed a generational leap in atmospheric science. Deterministic foundation models have proven capable of processing global states with unprecedented speed and spatial resolution, while generative and probabilistic diffusion models have effectively bridged the gap toward high-fidelity ensemble forecasting. Furthermore, domain-specific deep learning architectures—such as 3D U-Nets, Graph Neural Networks, and ConvLSTMs—are successfully addressing the highly non-linear, small-scale dynamics of localized severe weather, from tropical cyclone rapid intensification to flash droughts and convective squalls.

However, the transition from experimental superiority to operational infallibility is constrained by deep learning's intrinsic reliance on historical distributions. The inability of purely data-driven models to reliably extrapolate during unprecedented, record-breaking climate anomalies remains a critical vulnerability. Addressing this limitation requires a departure from standard error-minimization paradigms. Future advancements will depend on the widespread adoption of extreme-value loss functions, surgical fine-tuning techniques, and realistic, in-situ benchmarking frameworks. Most importantly, the path forward lies in physics-AI hybrid modeling, which fuses the computational efficiency and pattern-recognition capabilities of neural networks with the unbreakable thermodynamic constraints of physical laws, ensuring that early warning systems remain accurate, interpretable, and resilient in a rapidly changing global climate.

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